Games that Teach Project Plan

Team Members

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Notes and ideas

- Battle ship but you use addition and subtraction to shoot
- Over time it gets harder and harder
- Making it a map instead of a board
- 2 main levels plus one bonus if we can make it
- *** Mode one k-4 +&- ***
- Mode two 5-8 *&/
- Bonus level 9+ algebra
- Scales with time
- At the end of the level it will become word problems

Game Details

Game Title	Bunker Hunters
Game Concept	The player has to find all the hidden bunkers. The player chooses a square on a grid to fire at. Then the player has to solve math equations to lock in the coordinate. The player is trying to find all the bunkers and destroy them. The game ends when the player finds all the alien Bunkers.

Educational Goals

Learning Objective

Be better at math operations, specifically addition, subtraction, multiplication, and division.
Colorado state standards: 2.NBT.B. Number & Operations in Base Ten: Use place value understanding and properties of operations to add and subtract.

Practice

Solving math problems under a time crunch to hit their targets and win the game

Feedback

If they input different coordinates than what they clicked on then they lose one of their three lives. If they input the right coordinates then they hit the target and move on to the next stage or continue to try to find the enemies. If they get the first question wrong then they would know that they failed and they would lose a life.

Target Audience

Age Range / Grade Level

K-4, and 5-8

Background Knowledge

In theory how to use the math operators. They should also be able to just know simple addition and subtraction equations

Considerations

The attention span of the younger students is very short and we need to keep in mind that the older students need constant stimulus or else they will be very bored.

Core Gameplay

Gameplay Cycle

The is what the gameplay loop looks like:

Choose the target you want → do the 1st equation (what equals 7? 1. 3+3, 2. 3+4,3. 3+2)→ do the 2nd equation→ repeat

Player Controls

Player Action	Control	What Happens?
Move left/right	Left / right arrow keys	The targeting scope moves horizontally across the screen in the direction indicated.
Move up/down	Up / down arrow keys	The player moves vertically across the screen in the direction indicated
Choose box	enter	Chooses the bx that the user wants to destroy

Game Mechanics

Movement

They move the targeting scope to the box that they want to shoot and then they have to solve

equations to hit the right box

Scoring/Progress

The players know that they are winning by showing that the enemies are dying by showing one of the bunkers being blown up and having the user go on to the next wave of the level.

Challenge/Obstacles

They have to get both of the math problems correct to hit the target. However even if they hit the target the alien bunker might not be there so they have to keep shooting till they hit the target

Consequences

They lose a life and once they lose their starting three lives they lose the game.

Increasing Difficulty

Yes. The idea is that as they go on throughout the stage the math problems become more and more difficult ending with word problems.
Easier: 3+4
Harder: 17+5
Word problems: Jon has 18 apples and Sally ate 4 of his apples, how many apples does Jon still have?

Educational Mechanic

The equations for the user to find the coordinates of the enemies and get rid of them.

Gameplay

The game play has the following general feel at the beginning, middle, and end points.

Beginning	Easy math problems like 3+7
Middle	Harder math problems like 19-17 or 3+7-5
End	Word problems

Look/Feel

Setting / Theme

The game takes place in your secret base, where you learn that aliens have started to invade earth, fire missiles at their bunkers!

Visual Style

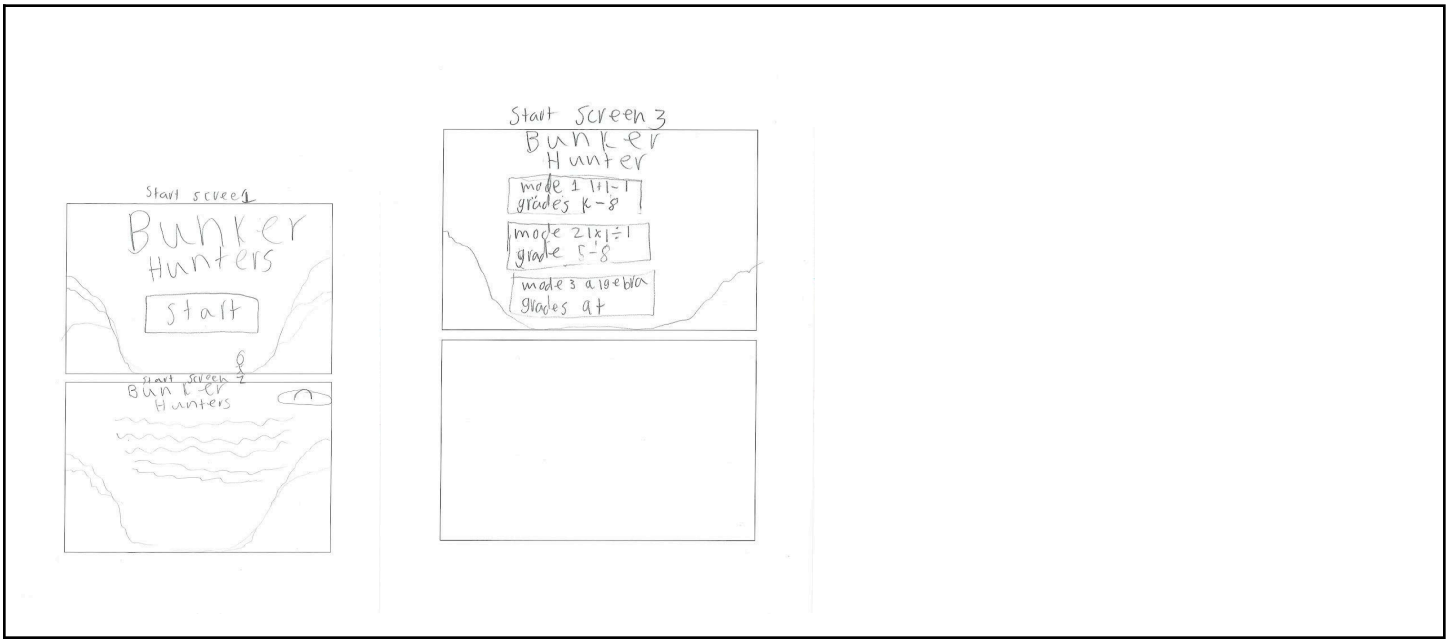
The game looks very simple, very basic, with maybe a background on the map

Mood

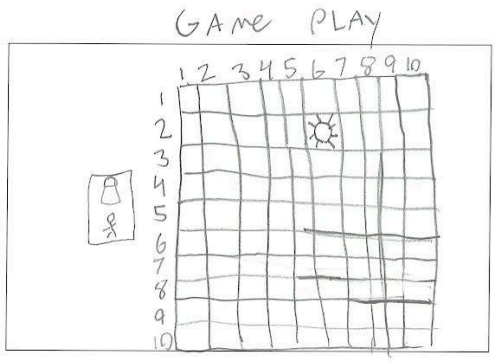
The player should feel excited and under pressure while playing the game.

Visuals

Start Screen



Game Play



Game Over



Revisions

Revision Description	How It Improved the Experience
